

Archita Garg

Software Engineer | C/C++ | Python | Networking | Testing & Automation | Full Stack

gargarchita02@gmail.com | +91 7497073843 | Yamunanagar, Haryana, India

linkedin.com/in/archita-garg-6a06b6288 | github.com/archita-garg02 | codolio.com/profile/ArchitaGarg02

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in C/C++, data structures and algorithms, object-oriented programming, and operating systems. Built C++ applications with file-based persistence, validation, debugging, and testing while applying Python fundamentals, Git, and software testing to support reliable delivery. Exploring full-stack development, Cisco wireless concepts, Wireshark-based tracing, and scripting-language automation with a focus on networking, test planning, security-minded development, collaborative problem solving, and cloud-ready feature validation.

TECHNICAL SKILLS

Programming: C++, C, Python, JavaScript

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Debugging, Testing, Validation, Problem Solving

Networking: Computer Networks Fundamentals, Networking Concepts, Client-Server Concepts, TCP/IP Fundamentals

Testing & Automation: Test Case Design, Functional Testing, Input Validation, Debugging, Test Automation Fundamentals

Development Tools: Git, GitHub, VS Code

Web & Mobile: React Native, React.js, HTML, CSS

AI & Emerging Technologies: Agentic AI, LangChain, LLM Applications

Core CS: Operating Systems, DBMS, Computer Networks

PROJECTS

Banking Management System

C++ | Object-Oriented Programming | File Handling | Testing & Debugging Developed a console-based banking management system in C++ to simulate core banking operations and account workflows. The application manages customer accounts and transactions while applying object-oriented programming principles, data validation, file-based persistence, and structured testing to improve reliability and maintainability. The project provided practical experience in designing application logic, handling invalid inputs, debugging functional issues, and maintaining consistent data across multiple operations.

- Engineered core banking workflows in C++ for account creation, deposits, withdrawals, balance enquiries, fund transfers, account search, and transaction history, strengthening access point-style transaction flow control and system reliability.
- Applied encapsulation, abstraction, inheritance, runtime polymorphism, and file handling to layer account logic, persist account and transaction records, and support maintainable object-oriented design.
- Designed and executed** test scenarios for invalid inputs, duplicate accounts, transaction constraints, minimum-balance requirements, and overdraft conditions while debugging, validating application behavior, and improving QA coverage.

TravelMate – Travel Planning Mobile Application

React Native | JavaScript | APIs | Git Developed a mobile travel-planning application using React Native to bring multiple travel-related services into a single user experience. The application includes authentication, weather information, maps, location-based services, cafes, rides, bookings, and user profiles. The project provided hands-on experience with reusable UI development, navigation, state management, asynchronous operations, API integration, input validation, testing, and debugging of mobile application workflows.

- Built multiple application workflows for authentication, weather, maps, cafes, rides, bookings, and user profiles using React Native and JavaScript, with a focus on digital user experiences and responsive team collaboration.
- Integrated** location search, weather information, and route navigation while implementing reusable components, navigation, state management, form handling, and asynchronous operations across the cloud-connected app experience.
- Validated and debugged core user workflows by checking inputs, reproducing functional issues, confirming fixes, and managing application source code with Git and GitHub for direct response to defects.

TRAINING & CERTIFICATIONS

Agentic AI Training

2026

Codenoid, Karnal

React Native Training

2025

Certiwise Learning Pvt. Ltd., Mohali

C++ Training

2024

Dell Soft, Yamunanagar

EDUCATION

Bachelor of Technology, Computer Science & Engineering

2023

Seth Jai Prakash Mukand Lal Institute of Engineering & Technology

8.11 / 10

Courses: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, DBMS